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Analysis of strain in cutting zone with FEM and stereological metallographic evaluation

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Abstract. Finite element modelling (FEM) of machining is a widely applied way to get information about the phenomena occurring during the cutting process. This paper discusses experimental work and FEM analysis to investigate the mechanism of CK45 (1.0503) carbon steel chip formation during orthogonal turning process. Local strain in cutting zone is estimated by measurement of deformation of metallographic cut using stereological evaluation of the grain boundary orientation. Estimated local strain in cutting zone was compared to deformation analysis for orthogonal cutting was made, based on simulation results in numerical modelling software DEFORM 2D. Stress, temperature, tool wear is also discussed in the last part of this paper.

Introduction

Machining is one of the oldest and important manufacturing approaches in the industrial world. Mechanism of the machining are an extremely complex process, has been subjected to systematic study for more than a century. To improve the efficiency of the machining process, the tool material should be selected carefully and also edge geometry must be designed precisely considering the optimization of cutting conditions. Lots of methods have been studied to select the appropriate parameters of machining. For example, the design of experiment in various methods [1]. Experimental work is time and cost consuming, the better way is modelling and simulation with validation experimental work. Modelling reduces experimental work considerably usually reduce errors.

The machining modelling is studied by many types of research in the word. In general for turning milling and drilling technology [2]–[4], however, FEM modelling and simulation can be used also for nonconventional machining such as laser assisted milling or laser cutting [5], [6]. The complexity of the phenomena machining processes is such that the computing time is too large. Although the computing power increases exponentially in a few decades, it is important simplified machining model to have acceptable computing time. Simplified 2D geometries orthogonal cutting and short simulated time (well below 1 mm cutting length) can reduce computing time from for several days, indeed several weeks to few minutes or a couple of hours [7].

Different methods can be used to accelerate numerical calculation. Basically, three type of analysis namely Eulerian, Lagrangian and the newer Arbitrary Lagrangian-Eulerian (ALE) can be used in FEM modelling of machining processes. In this paragraph, Lagrangian analysis has been used.

Modelling of orthogonal cutting

A thermo-mechanical finite element model was developed using DEFORM 2Dv11.0 SP1. It consists of the two-dimensional orthogonal cutting model in plane strain condition. The machined material is CK45 (1.0503) carbon steel.

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Geometry and boundary condition.

The machined workpiece is modelled like a rectangular box with the dimension of 1x5mm. The cutting model takes only account of the area of cutting zone and area near the face and rake of the tool. The experiment was used tool with plate SECO DNMG 150608-M3 Duratomic Ti(C1N)+AL_2O_3 technology coated grade. With respect to the cutting tool used in the experiment, the geometry of tool is modelled with a cutting edge radius of 30 µm, a first rake angle of -5° lengths of 0,15 mm, the second rake angle of 18° and flank angle of 7°.
Cutting parameters were used: feed f = 0,1 mm, cutting speed V_c = 175 m.min-1. Concerning the boundary condition, the workpiece is fixed. The under edge and the left edge of the workpiece are constrained against horizontal and vertical displacements. The tool moves from right to left towards the workpiece at the cutting speed. Initial geometry and boundary condition are presented in Fig. 1.

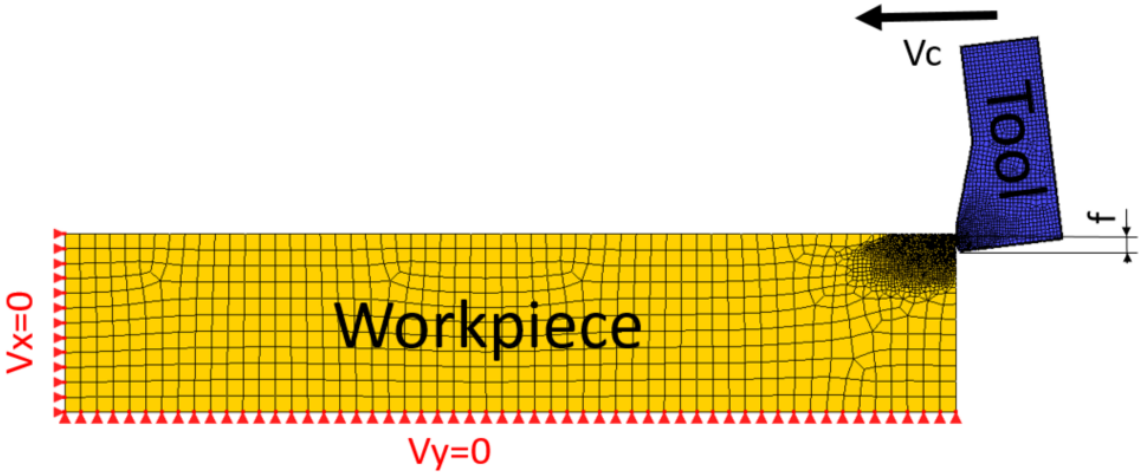

Fig. 1 Initial geometry and boundary condition

Behavior of the materials

The machined workpiece was considered plastic, the tool rigid. The plastic behavior of machined material data is described as a tabular data format. The tabular format is the most versatile format that can represent any material where flow stress can be given as a function of strain, strain rate, and temperature refer with: Eq. 1:

$$\bar{\sigma} = (\bar{\varepsilon}, \dot{\epsilon}, T) \tag{1}$$

As the workpiece material is considered isotropic, Von Mises' initial yielding criterion has been set according to [8]. Von Misses' yielding criterion describes: Eq. 2:

$$d\epsilon_{ij} = \frac{\partial f}{\partial \sigma_{ij}} d\lambda \tag{2}$$

Fig. 2. illustrates flow curve and points used to define the material behavior with defined temperature on the left and defined strain-rate on the right.

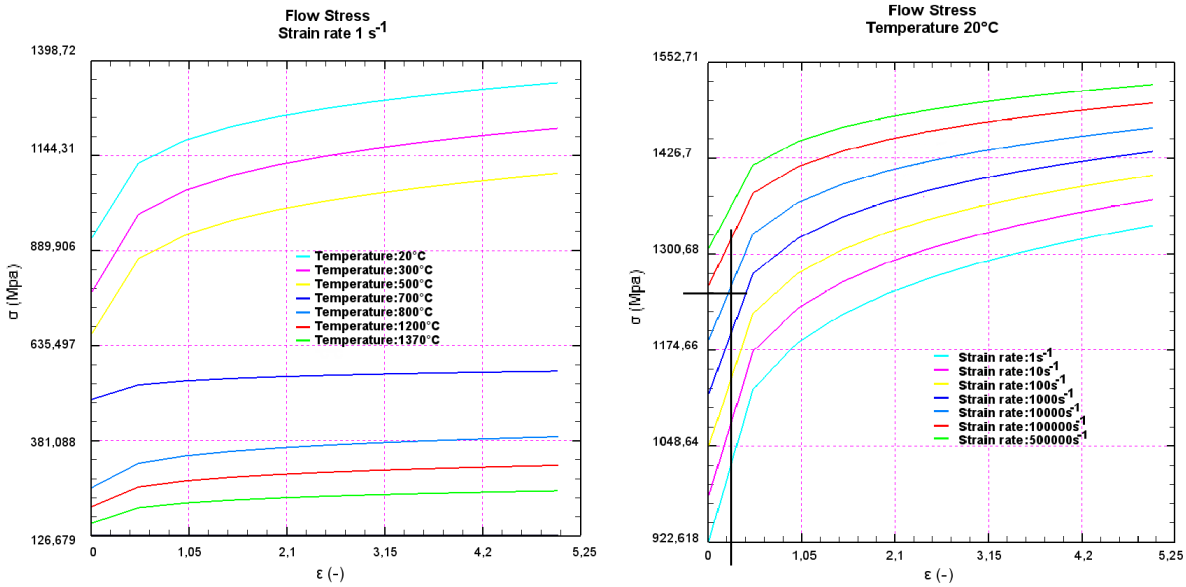

Fig. 2 Flow Stress curves of CK45 (1.0503) carbon steel

Tool material is carbide with DURATOMIC technology coated grade. The tool was considered rigid but for temperature array prediction tool geometry has meshed. Since the tool has been coated, tool mesh additionally contains two surface layers of TiCN and Al_2O_3 coating materials. The tool geometry with generated mesh can be seen in Fig. 3.

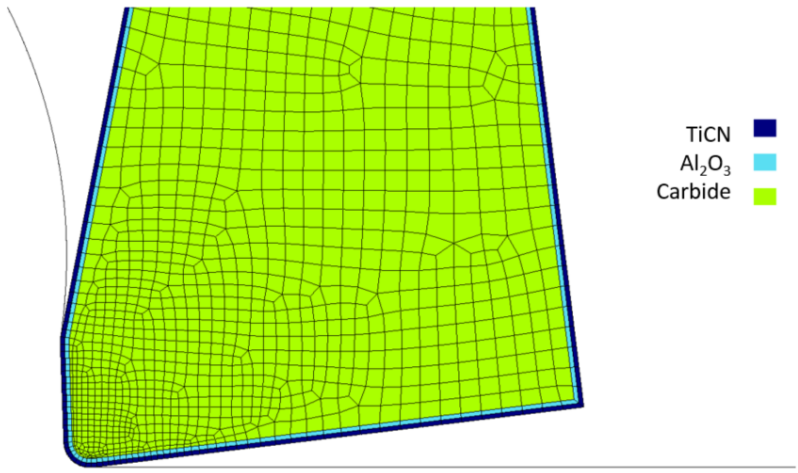

Fig. 3 Tool geometry and mesh

Tool thermal properties with is constant is presented the table 1. Thermal conductivity and Heat capacity of the Al_2O_3 are functions of temperature and it is presented graphically in Fig. 4.

Table 1. Al_2O_3 Heat capacity and Thermal conductivity

Thermal properties	TiCN	Al_2O_3	WC
Thermal conductivity ($Wm^{-1}K^{-1}$)	30	Fig. 4	59
Heat Capacity ($JKg^{-1}K^{-1}$)	15	Fig. 4	15
Emisivity(-)	0.2	0.3	0.3

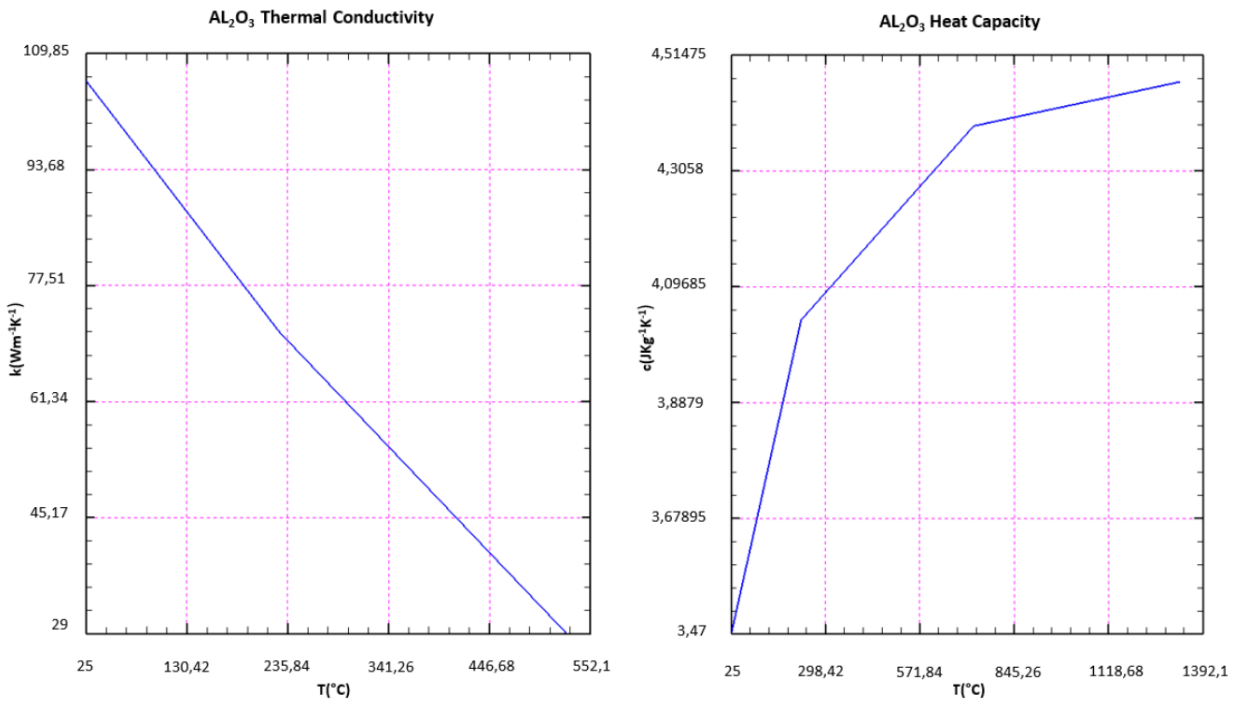

Fig. 4 AL_2O_3 Thermal Conductivity and Thermal Capacity

It is difficult to estimate friction at the tool-chip interface because of the very high normal pressure at the tool surface. For simplification of the numerical calculation is widely accepted that the tool-chip friction can be defined among the outer surface of the tool and all the nodes of the workpiece. Coulomb's friction with a low coefficient of value 0,05 in accordance with [9] is adopted to model.

Simulation settings

The convergence and the accuracy of the numerical procedure depend on the mesh of the workpiece. The size, number and type of the elements used in the mesh play a significant role in the simulation outcome as well. As a rule, a large number of small-sized elements increase accuracy but also increase computational time. Suitable meshing technique in machining has identified the region that is more interesting for the analyses. Finer meshes are used in regions of the primary and secondary deformation zone. In the region further away from the cutting edge a larger elements of the mesh were used. A side length finer mesh elements close to cutting edge is approximately of 0.004 mm. Side length of the larger element is approximately of 0.08 mm. Since the tool moves relative to the workpiece, it is necessary to ensure that the region where it is set a finer grid is shifted. Finer mesh in the region is always created new one when the re-meshing condition occurs. Continuous chip formation is simulated by changing the default re-meshing procedure. Chip formation is simulated by deleting the elements close to the tool cutting edge that has been subjected to high stresses. In order to simulation control, depends on the size of mesh elements and cutting parameters, constant $4{,}9.10^{-7}$ s/step increment has been set.

Result and discussions

For comparison of gain boundaries orientation on deformed experimental chip and FEM simulation result *Flownet* postprocessor function can be used. *Flownet* is a feature that can better visualize how the chip is formed during a simulation. *Flownet* is a pattern that can be applied to the cross-section of a workpiece which will be deformed. The postprocessor can have grid, polygon, offset and user defined *Flownet* pattern. For gain boundaries orientation in deformed places can be used polygon pattern where the size of polygons is approximately some like grain size. Experiment gain boundaries orientation and polygon *Flownet* pattern simulation are presented in Fig. 5.

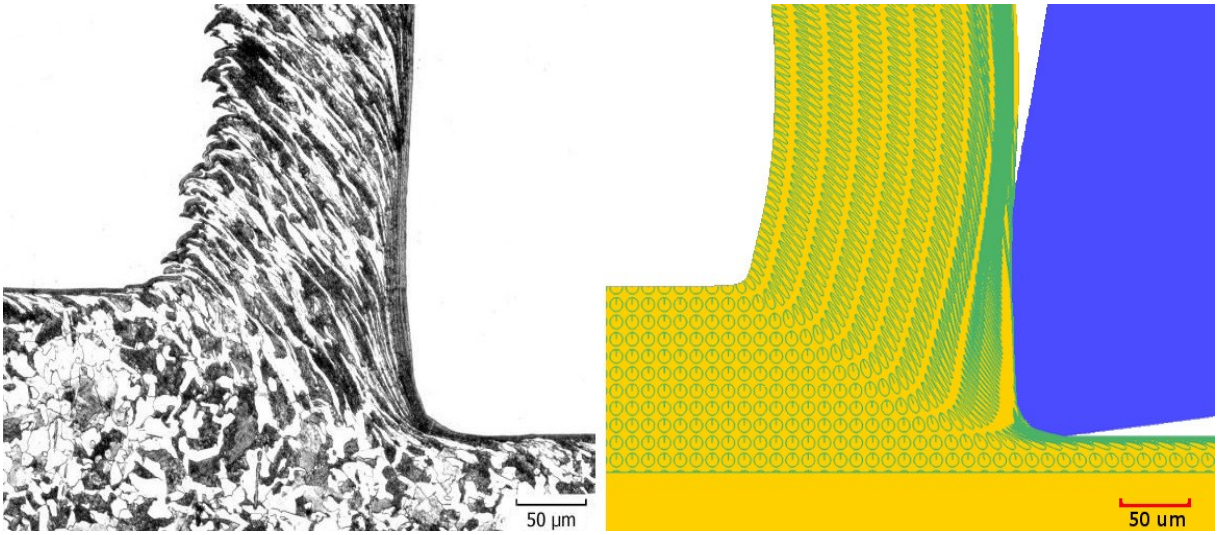

Fig. 5 Deformation in metallographic cut left, Flownet pattern simulation right

The same shear angle ϕ=30° was obtained both experimental result (see Fig. 5) [9] and in the FEM simulation. For better comparison graphically transformed polygon flow net pattern to metallographic cut can be seen in Fig. 6.

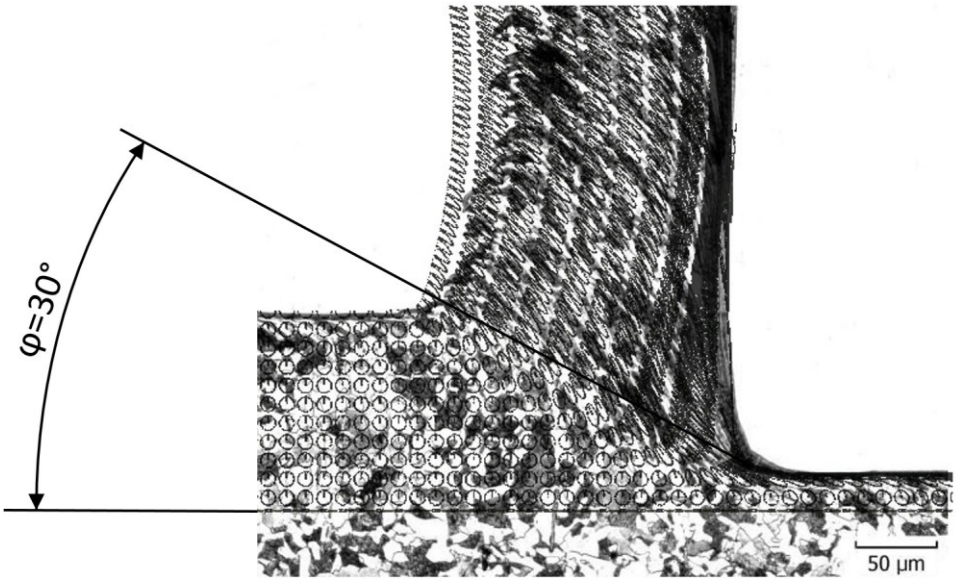

Fig. 6 Metallographic cut with simulation polygon pattern

Of course, FEM analysis provides more possibilities to explore stress-strain properties of chip formation. By focusing on primary cutting zone near the shear plane, we can determine max principal strain between 0,138 and 0,688 (B,C,D,E,F) as it is shown in the Fig. 7. Local plastic deformation was 0,63 in primary area using stereological evaluation [10]. From this result in case of strain rate of 1000 s^{-1} stress (estimated from flow stress curve at 20°C) is 1275 MPa. This result is in coincidence with numerical modelled one (see Fig. 9).

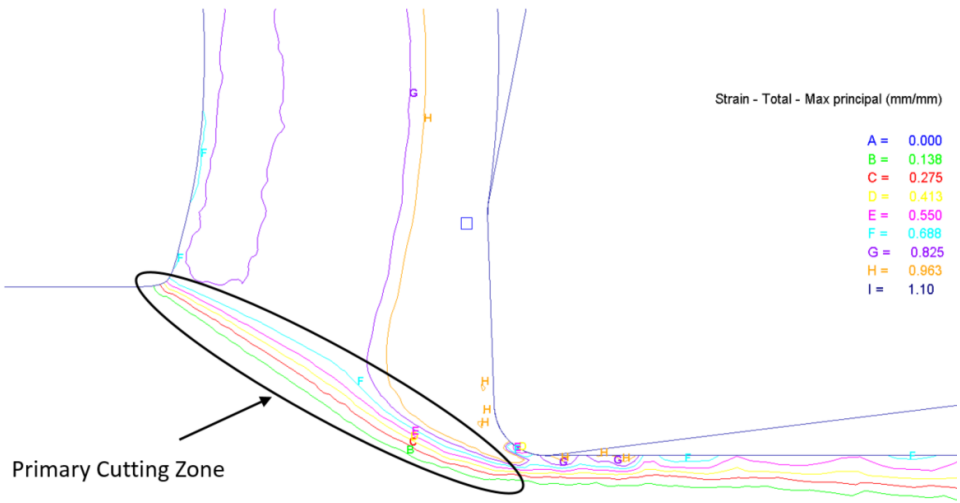

Fig. 7 Maximal principal strain in primary cutting zone

Fig. 8 presents component of cutting force obtained from FEM simulation. The graph shows that the cutting forces are stable after $1{,}5.10^{-4}$ s from the beginning of cutting simulation. After this time the chip formation is stable and the average size of cutting forces are $F_c = 222$ N and $F_t = 126$ N. Comparing the experimental results of forces measured with results in [11] due to the simplicity of orthogonal cutting simulation values of cutting forces are a realistic result. Morphology of chip formation after $1{,}5.10^{-4}$ and distribution of stable effective stress is shown in Fig. 9. The maximum effective stress 1260 MPa is located in the vicinity of the shear plane on the tooltip and in the small area on face plane of the cutting tool.

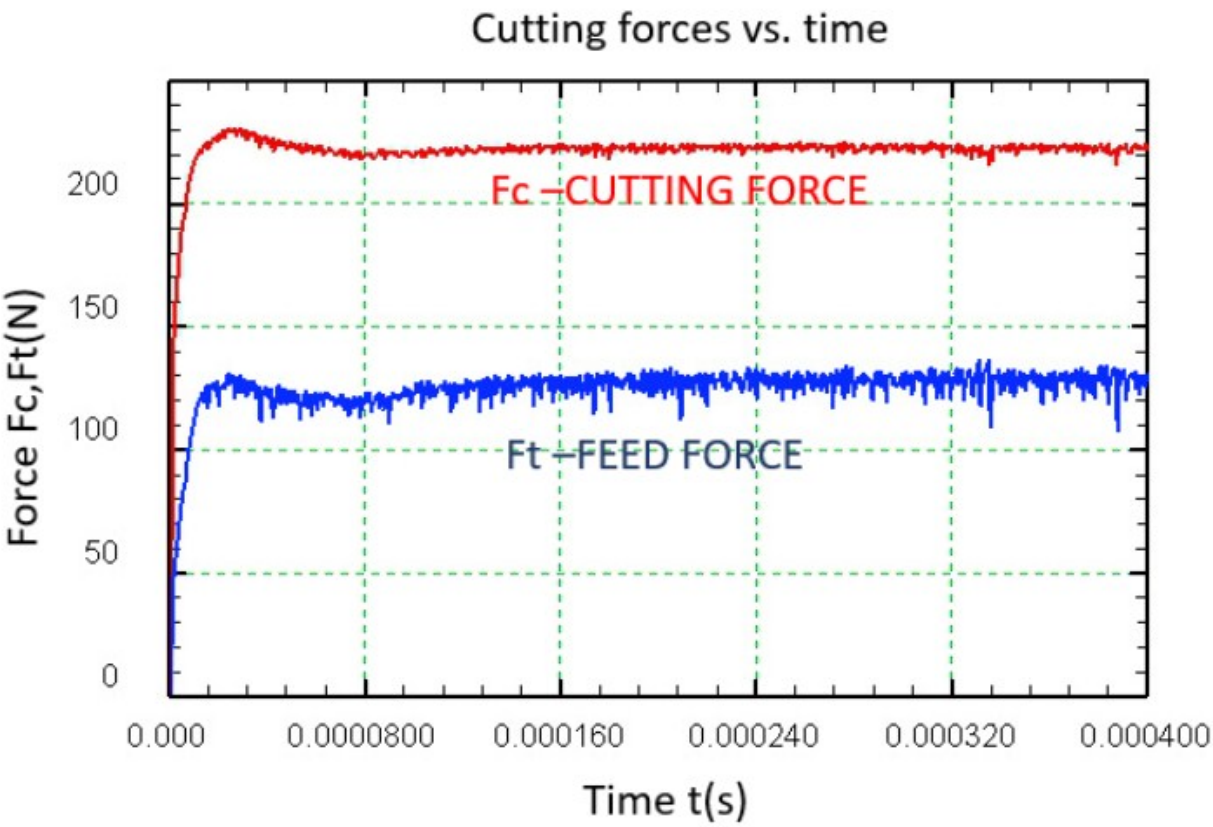

Fig. 8 Component of cutting force versus simulation time

Presented chip morphology and results obtained in the simulation is when the thermal softening factor is taken into account. It means that the material model and flow stress curves rake places between 20°C 1370°C, and thermal properties of workpiece material and cutting tool material are set correctly. Fig. 10 shows the temperature distribution during orthogonal cutting.

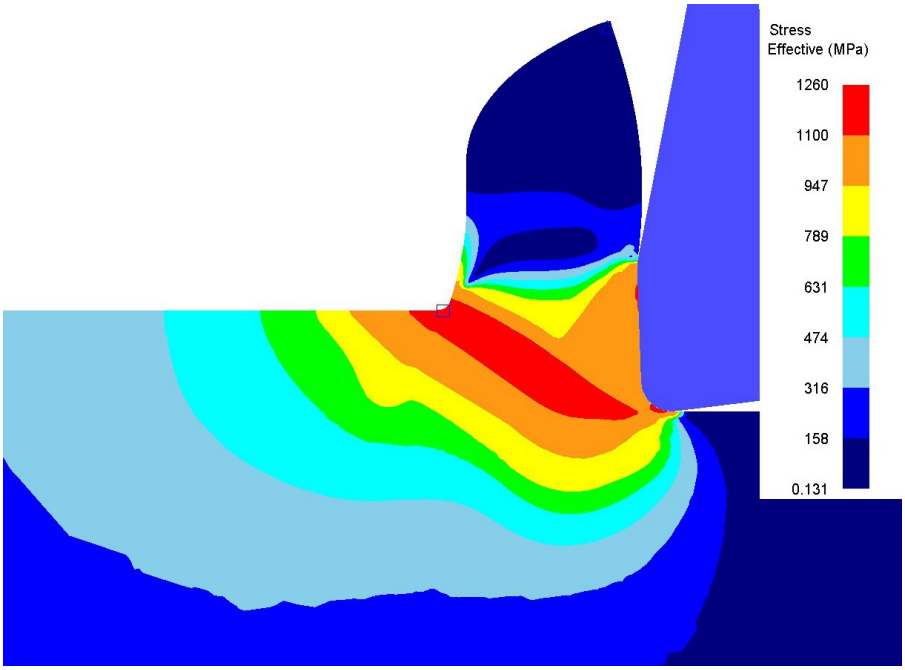

Fig. 9 Effective stress in cutting zone

It can be observed that most of the heat is kept in the chip. Heat transfer between tool and chip and friction heat generated make tool heated to a temperature of 410°C. Maximum chip temperature is 808°C. However, the final time simulation is $16{,}5.10^{-4}$s that is not enough time to predict tool temperature. On tool geometry was made two tracking points to examine temperature. As we can see on the graph the tool temperature rises.

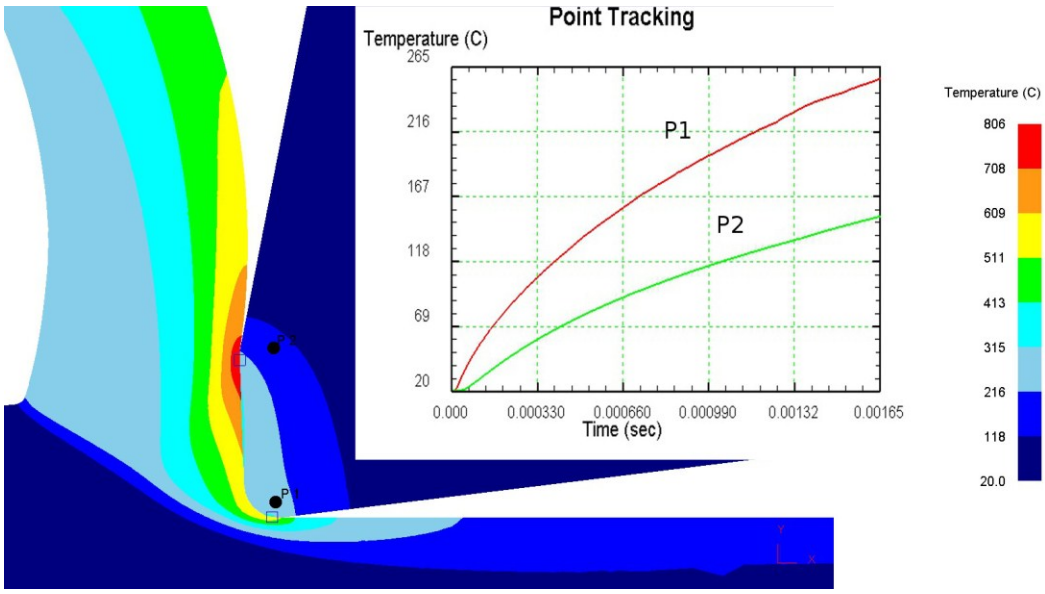

Fig. 10 Temperature cutting zone and tool distribution

Conclusion

In this study, 2D orthogonal cutting FEM simulation of turning have been developed. Prediction of 2D chip formation strains in cutting zone, cutting forces and temperature distribution has been investigated. FEM simulation of chip formation and strain in cutting zone was evaluated using the stereological metallographic cut method. The following conclusion can be drawn from the result of this study.

- Lagrangian 2D FEM method can be used for chip formation in orthogonal turning simulation.
- Simulation result depends on element size. Smaller size elements contribute to a more accurate result, but computing time is longer.
- The material model can by described as tabular data. An important condition is to use flow curves at different temperatures and at different strain-rate.

Acknowledgements

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